

Tsunghan Chiang

SOFTWARE · FIRMWARE · PERFORMANCE ENGINEER

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Summary

OS Engineer with 8 years in Linux development, performance optimization, and storage infrastructure. Dedicated to production reliability, team scaling, technical mentorship, and user-facing quality. Delivered NAS products driving 5K+ monthly sales at Ubiquiti and 190K+ software deployments at QNAP. Experienced in C, Go, Git, AI-aided tools, Debian, ZFS, Btrfs, Docker, Linux internals and benchmarking.

Work Experience

Ubiquiti Inc.

UNIFI-OS STORAGE SOFTWARE ENGINEER

Taiwan/Remote

Mar. 2022 - Present

- Built Ubiquiti's first NAS end-to-end --- core UX-aware Btrfs/ZFS middlewares, scaling to 6 models.
 - Developed across the UniFi-APPs, Debian OS, Linux kernel and BSP layers, collaborating for robust manufacturing and field deployment, guiding hardware strategy with benchmarking data.
 - Co-led testing frameworks with SQA team, implementing I/O stress loops, RAID expansion tests, and validation of BTRFS, ZFS and eCryptFS filesystems using xfstest and fio.
 - Established support checklists on performance and troubleshooting, handling 180+ escalations.
 - Upgraded Linux 4.19 to 5.10 along with drivers, unlocking OS capability and eBPF observability.
 - Fixed cross-layer issues from kernel (PHY crashes, NFS panics, AHCI IRQ, 93% socket memory) to storage (RAID 50% regression, FS write amplification, Samba socket leaks, Drive OOMs).
- Led full-stack system tunings and resource control across 6 NAS models.
 - Maximized data pipeline throughput (by 50%) via tuning CPU, IRQ, TCP and application layers.
 - Enhanced SSD-RAID BTRFS write IOPS by 40% via mq-deadline scheduler and ARM64 CRC32.
 - Built AI-skilled benchmarking platform spanning RAID, filesystem, and network stacks.
 - Applied cgroup V2 to prevent swap-induced latencies and prioritize foreground/background I/O.
- Drove system reliability, scalability and efficiency across all 2M+ UniFi-OS machines.
 - Architected ustated gRPC daemon, cutting CPU 65% and system load up to 50%.
 - Revamped the Debian build system for safe package upgrades with AI-aided workflows.
 - Conducted 280 code reviews annually and rolled out Claude PR-review CI across 17+ repos.

QNAP Systems Inc.

SOFTWARE ENGINEER

Taiwan

Jan. 2018 - Feb. 2022

- Led Hybridmount, a filesystem allowing local access to cloud files, reducing user-operating costs.
 - Built FUSE-based architecture with SQLite DB, libev event handling, and IPC.
 - Shipped multi-cloud mounting, client-side encryption, smart sync with predownload/autoupdate, cache pinning, and partial downloads across 190K+ devices.
 - Implemented CI pipelines using GitLab, Jenkins, and Docker with filesystem validation suites.
- Achieved 300% metadata, 50% I/O gains via multi-threading and caching.
 - Enabled VFS cache by handling unicode normalization, case-sensitivity and cache invalidation.
 - Resolved cyclic deadlocks between FUSE xattr and IPC daemons via API flow redesign.
 - Mitigated the inode limit and indexing bottleneck, scaling 50M+ files at 200 files/s.
 - Optimized download latency by 36% using centralized state machine with condition variables.
 - Reduced recursive deletion from 22s to 1s by batching syscalls and minimizing context switches.

Education

National Taiwan University

M.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

Taipei, Taiwan

Sep. 2014 - Jun. 2016

- GPA: 3.84/4.30
- Thesis: Program Analysis with A Loop-function-based Tracing Tool on Virtual Platform
 - Awarded Best Paper at the 2017 ACM Research in Adaptive and Convergent Systems.

National Chiao Tung University

B.S. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Hsinchu, Taiwan

Sep. 2010 - Jun. 2014

- Grade: 90.2% · Recipient of Academic Achievement Award for 4 semesters.

University of Illinois at Urbana-Champaign

EXCHANGE STUDENT IN COMPUTER SCIENCE

Champaign, IL

Aug. 2013 - Dec. 2013

Scholar Projects

Phase-based Profiling with Timing Approximate Simulators

CO-AUTHOR

- Achieved cycle-approximate simulation and to speed up design space exploration using QEMU.

Thread Cluster Memory on the UTAH Simulated Memory

AUTHOR

- Proposed a thread scheduling algorithm clustered by memory intensity, improving processing time by 10%, fairness by 19%, and energy efficiency by 21% vs. first-come-first-serve.

Leadership

Community Committee

CHAIRPERSON

Taipei, Taiwan

Jul. 2021 - Jun. 2023

- Streamlined admin tasks and reduced successor ramp-up via reusable templates and handover procedures.
- Optimized community regulations regarding meeting quorum and the election system.

EECS Student Council

PRESIDENT

Hsinchu, Taiwan

Sep. 2012 - Jun. 2013

- Organized after-school programming tutoring for 6 struggling students.
- Hold an orientation event, connecting freshmen with professors and senior students.